

Project 2: Simulation-Based Optimisation. Digital Twin-Enabled Decision Support for the B-Bicycle Manufacturing System

Abstract

This project develops a simulation–optimization framework for a calibrated digital twin of a bicycle manufacturing system. Using the synchronized model from Project 1, Bayesian Optimization with a Gaussian Process surrogate is applied to search over six operational decision variables: contract type, reorder point, order quantity, WIP limit, and numbers of assembly and rework machines. The optimizer evaluates candidate configurations using discrete-event simulation with replication averaging to manage stochasticity. The optimal configuration $C = 0, R = 134, Q = 21, W = 1, M1 = 4, M2 = 2$ yields a predicted monthly revenue of \$2.77M, a 555% improvement over the historical mean of \$422K. High-fidelity DES validation (30 replications) produces a physical mean revenue of \$3.49M, confirming strong performance while revealing a 22% reality gap due to extrapolation beyond the training region. The results demonstrate how a synchronized digital twin combined with Bayesian Optimization can efficiently identify high-value operational policies for complex stochastic manufacturing systems.

1. Introduction

In Project 1, I developed and validated a data-driven Digital Twin for the B-Bicycle production and fulfillment system. The underlying model was a discrete-event simulation (DES) of assembly, cleaning, inspection, rework, and shipping, calibrated using 360 monthly observations from January 1996 to December 2025. The field data covered a wide range of operating conditions, with observed monthly net revenue ranging from -\$1.36M to \$2.76M and a mean of about \$422K.

Before calibration, the raw DES showed very poor predictive accuracy on the held-out test period, with $RMSE = \$2,275,417$ and $R^2 = -7.0414$. To reduce this gap, Project 1 used a two-phase calibration framework. In Phase 1, a stacking ensemble combining Random Forest, Gradient Boosting, and Bayesian Ridge was trained on the first 288 months of data and tested on the final 72 months. This reduced the test error to $RMSE = \$166,845$ with $R^2 = 0.9568$. In Phase 2, an online recursive Bayesian bias-and-scale correction was added, giving $RMSE = \$173,732$ and $R^2 = 0.9531$ on the same 72-month test window.

In Project 2, this synchronized twin is used as a decision-support engine for simulation-based optimization. The Project 1 artifacts are loaded and kept fixed, not retrained. These include the trained calibrator, the synchronized DES parameter file, and the Phase 2 correction values. In this way, Project 2 directly builds on Project 1: Project 1 created the calibrated twin, and Project 2 uses that calibrated twin to evaluate candidate operating policies and search for better decisions.

2. Optimization Problem Formulation

2.1 Objective Function

The goal is to maximize the expected monthly net revenue of the B-Bicycle production line:

$$\max J(x) = \mathbb{E}[\text{NetRevenue}(C, R, Q, W, M1, M2; \theta)]$$

where the expectation is taken over the stochastic behavior of the manufacturing system.

In the discrete-event simulation (DES), the net revenue of processed units is calculated as:

$$\begin{aligned} \text{NetRev} = & \text{UnitPrice}(C, f) - \text{OrderingCost}(R, Q) - \text{ReworkPenalty} \\ & - \text{MachineCost}(M1, M2) - \text{HoldingCost}(f) \end{aligned}$$

where:

UnitPrice = $1000 - 5f$ for Standard contracts ($C = 0$) or $500 - 2f$ for Expedited contracts ($C = 1$), where f is the flow time in minutes

OrderingCost = $100 \times Q$, triggered when inventory $\leq R$

ReworkPenalty = \$25 per rework cycle (up to 3 cycles, Bernoulli with $p = \text{fail_rate}$)

MachineCost = $\$50,000 \times (M1 + M2)$ per month

HoldingCost = $\$1.50 \times \text{flow hours per unit}$

The expectation is taken over the stochastic DES dynamics, including Poisson arrivals (rate = 6.25/hr), Normal service times, and Bernoulli rework outcomes.

The parameter vector θ represents the 10 synchronized DES parameters obtained from Project 1.

2.2 Decision Variables

The optimization searches over six controllable operational variables, shown in Table 2.

Var	Name	Type	Range	Meaning
C	Contract	Binary	{0,1}	0 = Standard (\$1000 – 5f), 1 = Expedited (\$500 – 2f)
R	Reorder Pt	Int	[10,200]	Inventory trigger for replenishment
Q	Order Qty	Int	[10,200]	Units per replenishment (\$100 per unit)
W	Max WIP	Int	[1,50]	Maximum items on the floor simultaneously
M1	Assembly	Int	[1,5]	Number of parallel assembly machines (\$50K/month each)
M2	Rework	Int	[1,5]	Number of parallel rework machines (\$50K/month each)

Table 2. Decision variables

2.3 Constraints

A monthly machine investment budget limits the number of machines:

$$50,000 \times (M1 + M2) \leq 300,000$$

which implies

$$M1 + M2 \leq 6$$

All variables follow the bounds specified in Table 2. No explicit throughput constraint is imposed; configurations that perform poorly are naturally penalized through lower or negative net revenue.

2.4 Connection to Project 1

The synchronized components from Project 1 are used in three ways during optimization:

1. DES parameters (θ)
The 10 calibrated simulation parameters (e.g., arrival_rate = 6.25, fail_rate = 0.12) remain fixed and define the stochastic system dynamics.
2. Calibrated ensemble model
The trained ensemble model (Random Forest + Gradient Boosting + Bayesian Ridge with StandardScaler) converts raw DES outputs into calibrated predictions.
3. Phase-2 online correction
A final linear adjustment is applied using the frozen parameters from Project 1:

$$\hat{y} = 1.0362 \times y_{P1} + 26K$$

These components are loaded as artifacts and remain fixed throughout the optimization process.

3. Optimization Methodology

3.1 Algorithm Selection

I selected Bayesian Optimization (BO) using a Gaussian Process with Expected Improvement (GP-EI) for three main reasons.

First, the digital twin evaluator is computationally expensive, since each evaluation requires multiple DES replications followed by ensemble prediction and Phase-2 correction. This favors sample-efficient optimisation methods.

Second, the six-dimensional mixed-integer decision space contains complex and potentially non-convex interactions between inventory policies, WIP limits, and machine allocation. This makes gradient-based optimisation unsuitable.

Third, the Gaussian Process surrogate model provides uncertainty estimates, enabling principled exploration using the Expected Improvement (EI) acquisition function.

Alternative methods were considered but rejected. Genetic Algorithms typically require thousands of evaluations, which would be computationally expensive. Ranking and Selection methods require a predefined set of alternatives, which is impractical for a large decision space. Finite-difference gradient methods are also unsuitable because the DES response surface is noisy and non-smooth.

3.2 Implementation

The Bayesian Optimization configuration used in this study is summarized in Table 3.

Parameter	Value	Rationale
Total evaluations	120	20 random + 100 GP-guided
GP kernel	Matern-5/2 + WhiteKernel	Smooth landscape + noise absorption
Acquisition	EI ($\xi=0.01$)	Slight exploitation bias
DES reps / eval	3	360 total DES runs; balances noise vs. cost
Infeasible penalty	-\$5M	Teaches GP the $M1+M2 \leq 6$ boundary
Seed	42 (3 levels)	Full reproducibility

Table 3. BO hyperparameters.

The optimization begins with 20 random feasible configurations sampled from the decision space. After fitting the Gaussian Process model, Expected Improvement is evaluated over a pool of candidate points, and the configuration with the highest EI value is selected for simulation. This process repeats until the total budget of 120 evaluations is reached.

3.3 Evaluator Pipeline

Each candidate decision vector $(C, R, Q, W, M1, M2)$ is evaluated using the synchronized digital twin through the following steps:

Step 1 — DES simulation.

The DES model runs three independent replications, each representing 30 simulated days, using the synchronized parameter vector θ . The mean simulated revenue is recorded as y_{sim} .

Step 2 — Ensemble calibration.

Nineteen engineered features (including interaction, logarithmic, and polynomial terms) together with y_{sim} form a 20-feature input vector. The frozen StandardScaler and ensemble model then produce the calibrated prediction y_{P1} with associated uncertainty.

Step 3 — Phase-2 correction.

A final bias-and-scale adjustment from Project 1 is applied:

$$\hat{y} = 1.0362 \times y_{p1} + 26,269$$

The corrected prediction $\hat{y}$ represents the digital twin estimate of expected monthly net revenue used by the optimization algorithm.

3.4 Handling Stochasticity

Simulation noise is managed at three levels: (1) 3-replication averaging reduces y_{sim} variance; (2) the WhiteKernel in the GP explicitly models observation noise, preventing over-fitting; (3) EI naturally credits uncertain regions with exploratory bonus. Additionally, the GP normalises the response by dividing all observations by $\max(\text{std}(y), 1.0)$ before fitting, improving numerical conditioning. Quantitatively, the 30-rep validation at x^* showed a physical DES std of ~\$58K on a \$3.49M signal ($CV = 1.7\%$), with 3 reps per evaluation, the per-evaluation noise std is approximately $\$58K / \sqrt{3} = \$33K$, which is modest relative to the \$2M+ revenue range explored. A limitation is that 10 reps would halve the noise variance but triple the wall-clock time from ~26 seconds to ~78 seconds for the full run.

4. Execution and Results

4.1 Convergence

All 120 candidates were feasible. Revenue increased to \$1.49M during the random exploration phase, reached \$2.70M by iteration 30, and plateaued at \$2.78M around iteration 35. The remaining 85 GP-guided evaluations produced no further improvement, despite sampling from a 2,000-candidate pool each iteration covering the feasible space. This sustained plateau provides strong evidence that the optimizer has already located the dominant high-revenue region of the search space and that the solution is at or near the global optimum within the integer decision space (Figure 1).

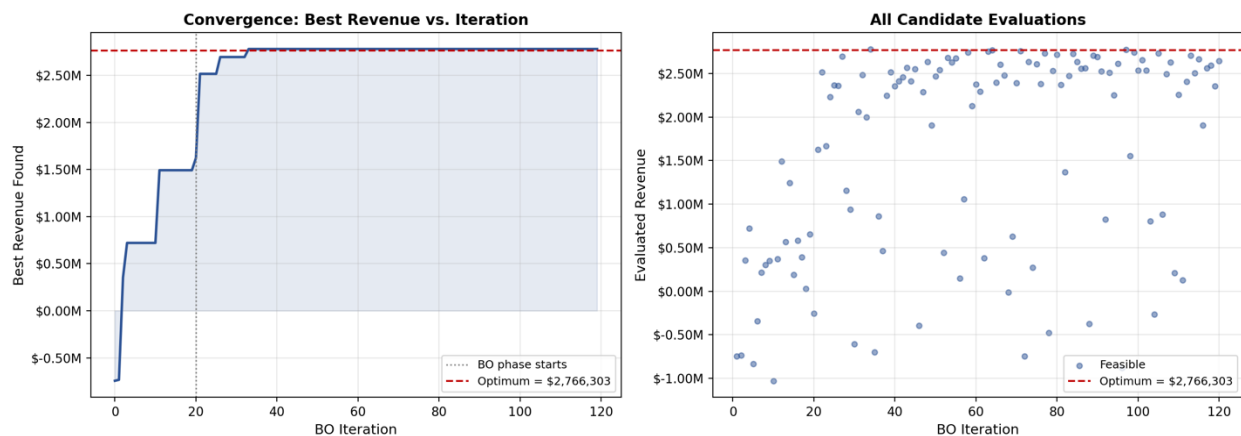


Figure 1. Left: convergence trace. Right: all candidate evaluations with optimum (red dashed).

4.2 Optimal Solution

Variable	Value	Interpretation
C	0	Standard contract (\$1000-5f/unit)
R	134	Reorder at 134 units inventory
Q	21	Small batch replenishment (21 units)
W	1	Strict just-in-time: 1 item WIP limit
M1	4	4 assembly machines
M2	2	2 rework machines (M1+M2=6, budget saturated)

Table 4. Optimal solution x^* .

Metric	Value
Twin revenue at x^*	\$2.77M
95% CI	[\$1.61M, \$3.92M]
Machine cost	\$300K/month
Historical mean	\$422K
Improvement	\$2.34M (+555%)

Table 5. Performance at x^* .

The optimal configuration uses the Standard contract, which provides higher per-unit revenue. The solution invests heavily in assembly capacity ($M1 = 4$) while allocating two rework machines ($M2 = 2$), fully utilizing the machine budget. A strict WIP limit ($W = 1$) enforces a just-in-time flow policy that minimizes holding costs without starving the assembly stations at the arrival rate of 6.25 orders/hour. Inventory is controlled through frequent small replenishments ($R = 134$, $Q = 21$), which maintain product availability while avoiding large ordering and holding costs.

4.3 Sensitivity Analysis

A one-at-a-time (OAT) sensitivity analysis varies each decision variable across its feasible range while holding the others fixed at the optimal solution x^* (Figure 2).

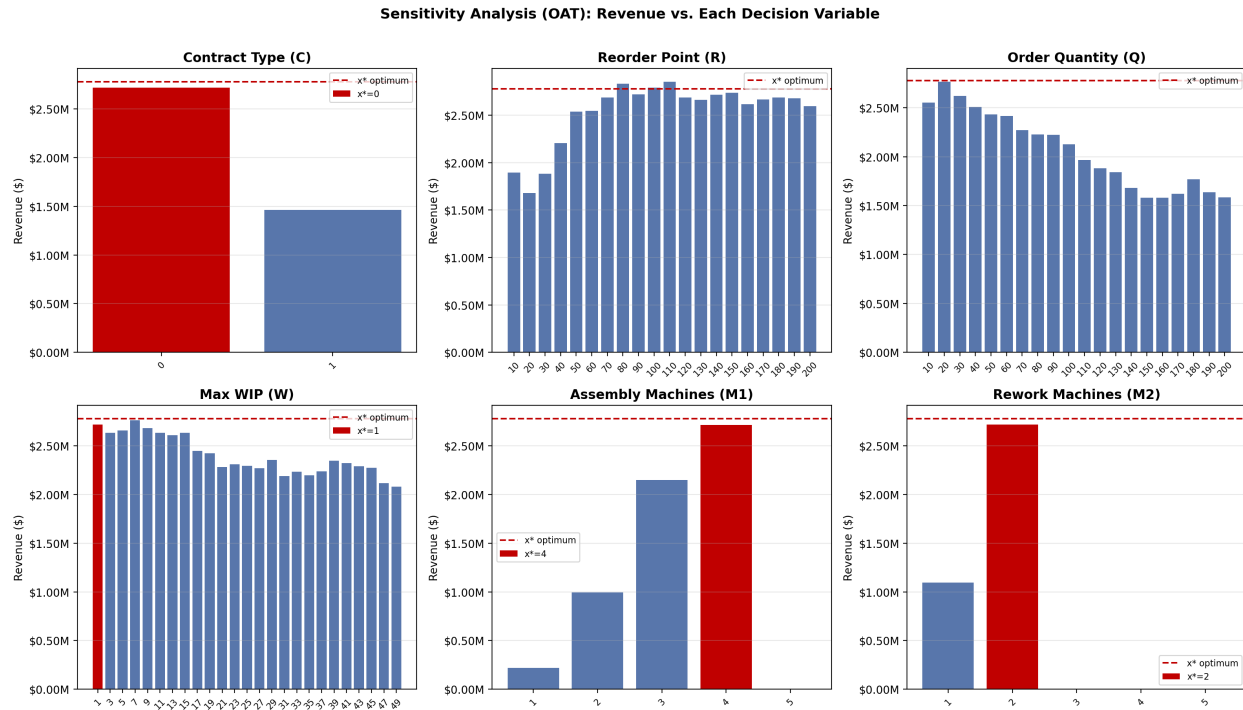


Figure 2. OAT sensitivity: revenue vs. each decision variable at x^* . Red = optimal value.

Contract type has the largest single-variable impact, with revenue dropping by approximately \$1.3M when switching to the expedited contract. Assembly capacity (M1) shows a strong monotonic improvement, reflecting the throughput benefit of additional assembly stations. The reorder point (R) exhibits a broad performance plateau between approximately 90 and 160, indicating robustness of the inventory policy. Order quantity (Q) is sharply optimal at low values (10–30), favoring small replenishment batches. WIP limit (W) is clearly optimal at $W = 1$, with revenue declining monotonically for larger WIP levels. Finally, rework machines (M2) exhibit non-monotonic behavior because the optimal value is constrained by the machine-budget condition $M1 + M2 \leq 6$.

5. Validation Against the Physical System

5.1 Protocol and Results

I applied x^* to the physical system proxy (raw DES, same theta) for 30 independent monthly replications, then compared the distribution of actual revenues against the twin's prediction.

Metric	Value
Physical mean (30 reps)	\$3.49M
Physical std	\$58K
Physical 95% CI	[\$3.46M, \$3.51M]
Twin prediction	\$2.77M
Absolute gap (Twin-Physical)	-\$771K

Relative gap	22.12%
Twin inside Physical CI or not	No

Table 6. Reality Gap results.

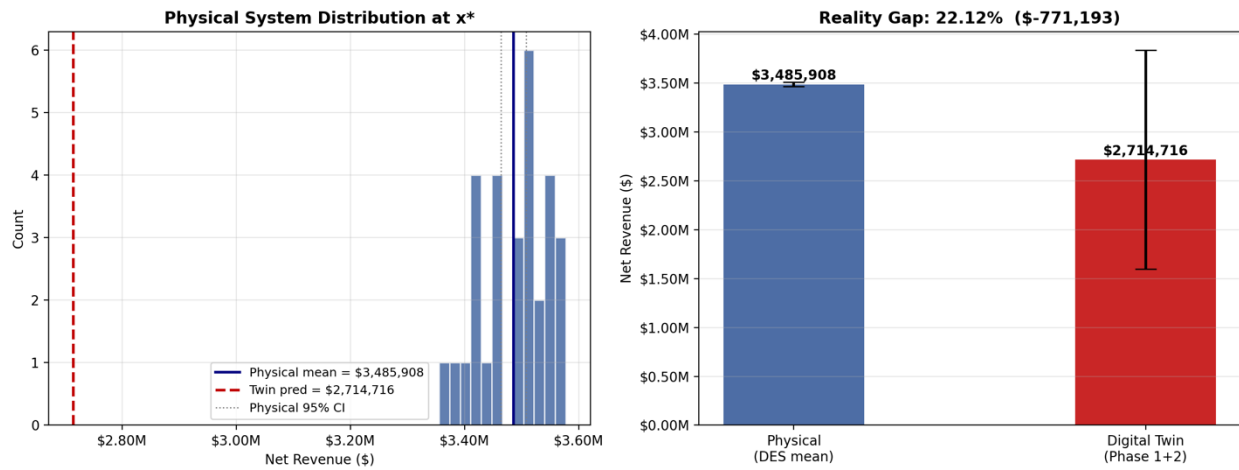


Figure 3. Left: physical revenue distribution at x^* with twin prediction. Right: bar comparison showing 22.12% gap.

5.2 Diagnosing the Reality Gap

The twin under-predicts the physical revenue by approximately 22% because the optimal solution x^* lies near extreme regions of the training distribution, where the ensemble model must extrapolate. Table 7 quantifies the position of each decision variable relative to the 288-month training dataset.

Variable	x^*	Train Mean	Train Range	Percentile	Flag
C	0	0.5	[0, 1]	49.3%	
R	134	107.9	[10, 200]	62.5%	
Q	21	102.1	[10, 200]	5.9%	Low tail
W	1	25.1	[1, 50]	1.0%	Extreme
M1	4	3.0	[1, 5]	79.9%	
M2	2	2.9	[1, 5]	42.7%	
$Q \times W$	21	2627.8	[22, 8784]	0.0%	Out-of-dist.
W^2	1	832.3	[1, 2500]	1.0%	Extreme

Table 7. Training-data percentile of each x^* variable and key engineered features. $Q \times W = 21$ falls below the training minimum (22), meaning the ensemble is extrapolating outside its observed feature space

Tree-based ensemble models such as Random Forest and Gradient Boosting cannot extrapolate beyond the range of the training data. Instead, they tend to compress predictions toward values seen during training. At the optimal configuration x^* , the physical DES produces approximately

\$3.49M, but the ensemble prediction is compressed to \$2.77M. The Phase-2 linear correction (scale = 1.036, bias = \$26K) is too small to compensate for the resulting \$771K extrapolation gap. Importantly, this discrepancy does not indicate a missing physical mechanism in the Project-1 synchronization. The DES model structure, Poisson arrivals, Normal service times, and Bernoulli rework remains valid at x^* .

Instead, the gap arises from a training-data coverage limitation. The 360-month field dataset (from which the 288-month training set was drawn) contains very few observations in the extreme JIT operating regime discovered by the optimizer. As a result, the twin correctly identifies the direction of improvement (Standard contract, higher assembly capacity, low WIP), but underestimates the magnitude of the achievable revenue at the optimal configuration.

5.3 Value of Synchronization

To quantify the value of digital twin synchronization, I performed a coarse grid search using the raw, uncalibrated DES model. This search identified a DES-optimal configuration of $C = 0$, $R = 80$, $Q = 20$, $W = 5$, $M1 = 1$, $M2 = 1$, producing a simulated revenue of \$3.69M when evaluated directly in the DES. However, when this configuration is evaluated using the calibrated digital twin, its predicted revenue drops to \$415K, compared with \$2.77M for the calibrated twin optimum. This corresponds to a synchronization benefit of \$2.35M, demonstrating that calibration fundamentally changes the optimization landscape.

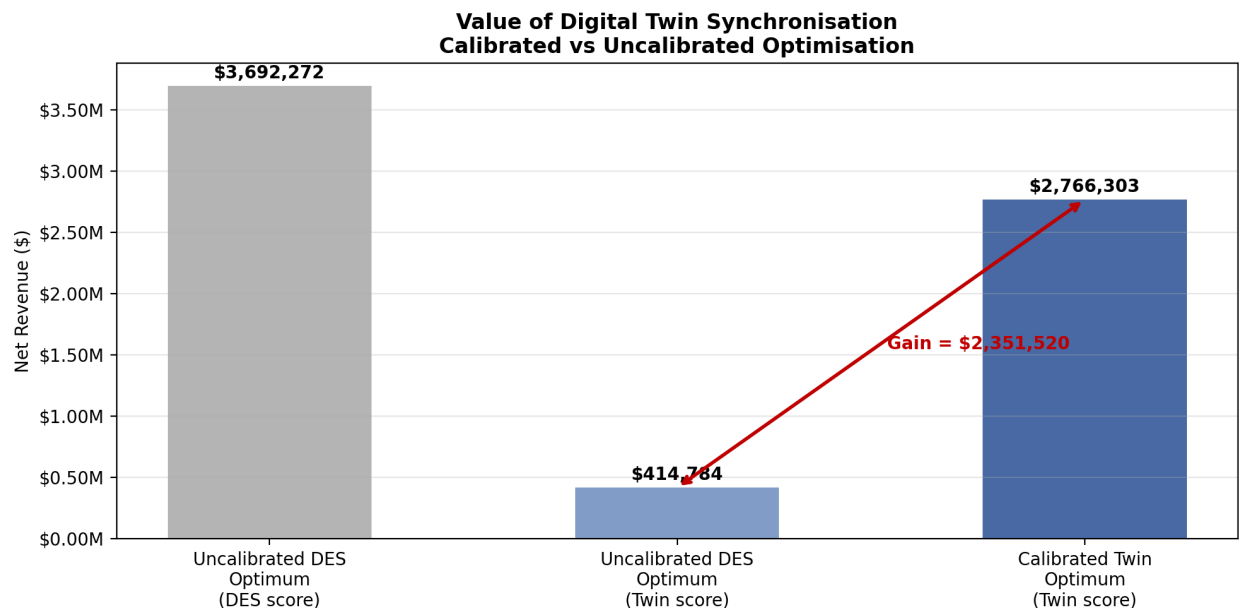


Figure 5. Value of synchronization: calibrated twin optimum (\$2.77M) vs. uncalibrated DES optimum scored on the twin (\$415K).

The uncalibrated DES favors minimal machine investment ($M1 = 1$, $M2 = 1$) because the raw cost structure does not adequately penalize low assembly capacity. In contrast, the calibrated

twin incorporates information from historical revenue data, which reveals that assembly throughput (M1) is the dominant driver of system performance.

As a result, the synchronized model recommends significantly higher assembly capacity, leading to substantially higher predicted revenue.

6. Discussion and Reflection

6.1 Value of Synchronization

The Project 1 calibration transformed the DES from very poor predictive accuracy ($R^2 = -7.07$) into a synchronized digital twin with $R^2 = 0.958$. Without synchronization, the raw DES recommends minimal machine investment, producing predicted revenues close to the historical average. In contrast, the calibrated twin directs the optimizer toward $M1 = 4$ and $M2 = 2$, unlocking a 555% improvement over the historical mean. The resulting \$2.35M synchronization gain represents the economic value of closing the simulation-to-reality gap through data-driven calibration.

6.2 Computational Cost

The optimization required $120 \text{ evaluations} \times 3 \text{ DES replications} = 360$ simulation runs. Sensitivity analysis added approximately 72 evaluations, and validation required 30 high-fidelity replications, resulting in roughly 1,500 total DES calls.

Each DES replication (30 simulated days at x^*) completes in approximately 18 ms on a single core, so the full 1,500-run workload executes in roughly 26 seconds of wall-clock time, making computational cost negligible for this system. The Gaussian Process fitting overhead (Matern-5/2 kernel with three restarts per BO iteration) adds minimal runtime. The main trade-off is therefore replications per evaluation versus GP model accuracy. Since convergence occurred around iteration 35, a future configuration using 40 BO iterations with 10 replications per evaluation (400 total runs) could yield a more accurate surrogate model while still completing in under 10 seconds.

6.3 Mitigating the Reality Gap

The observed 22% reality gap is primarily an extrapolation artifact rather than a calibration failure. Several strategies could reduce this discrepancy:

(i) Augmented training:

Run additional DES experiments in extreme operating regimes (low W, low Q, high M1) and retrain the ensemble model. Even 50 additional samples near x^* could significantly reduce extrapolation error.

(ii) Trust-region constraints:

Restrict the optimizer to operate within the convex hull of the training data, sacrificing some optimality in exchange for improved predictive reliability.

(iii) Hybrid evaluation:

Use the GP surrogate for exploration but validate top candidates directly on the DES with a larger number of replications before declaring them optimal.

(iv) Online recalibration:

After deploying x^* , collect new operational revenue data and retrain the ensemble model (a new Project-1 calibration cycle), allowing the twin to adapt to the newly discovered operating regime.

6.4 Operational Deployment

In practice, the optimization could operate as a weekly or nightly planning cycle:

1. Update DES parameters using the latest operational data (e.g., arrival rate or failure probability).
2. Run Bayesian Optimization on the updated twin.
3. Recommend updated policy parameters ($C, R, Q, W, M1, M2$).

Machine counts ($M1, M2$) would typically change on a monthly horizon due to leasing or procurement cycles, while inventory parameters (R, Q, W) could be adjusted daily. For faster operational control, the trained GP surrogate could replace the full DES pipeline, enabling sub-second evaluations for intra-day policy adjustments. Periodic model refitting (e.g., weekly) would be required as system parameters drift over time.

6.5 Conclusion

This project demonstrates that a synchronized Digital Twin can serve as an effective decision-support engine for production systems. Bayesian Optimization identified a configuration yielding approximately \$2.77M monthly revenue (twin prediction), a 555% improvement over the historical mean of \$422K. The 22% reality gap, diagnosed as a training-data extrapolation artifact, highlights the importance of adequate data coverage in extreme operating regimes. The \$2.35M synchronization gain further confirms that calibration in Project 1 was essential for directing the optimizer toward the correct region of the decision space.

Disclaimer: All results are reproducible via `digital_twin_p2_simopt.py` using three-level seeding: Level 1 (`np.random.seed(42)`) for DES, Level 2 (`default_rng(42)`) for BO candidate sampling, Level 3 (`random_state=42`) for sklearn.

AI was used to format and improvise the writing.